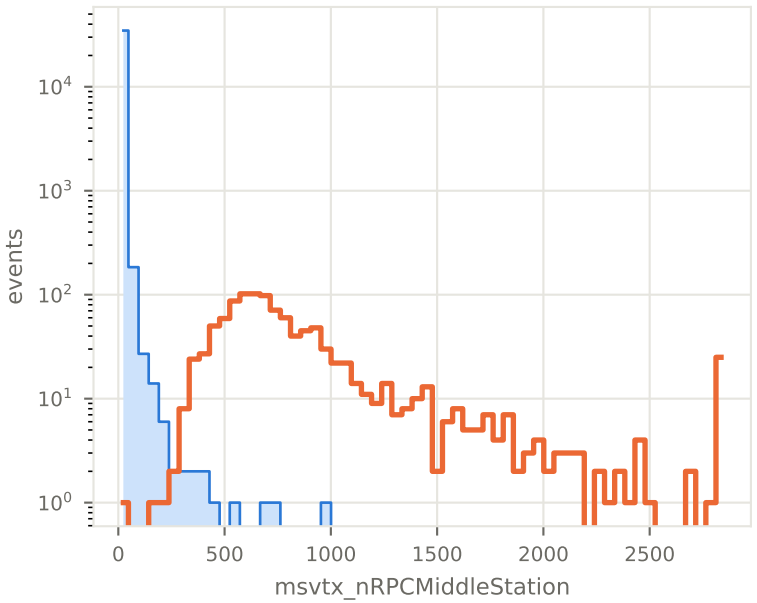
# msvtx_nRPCMiddleStation

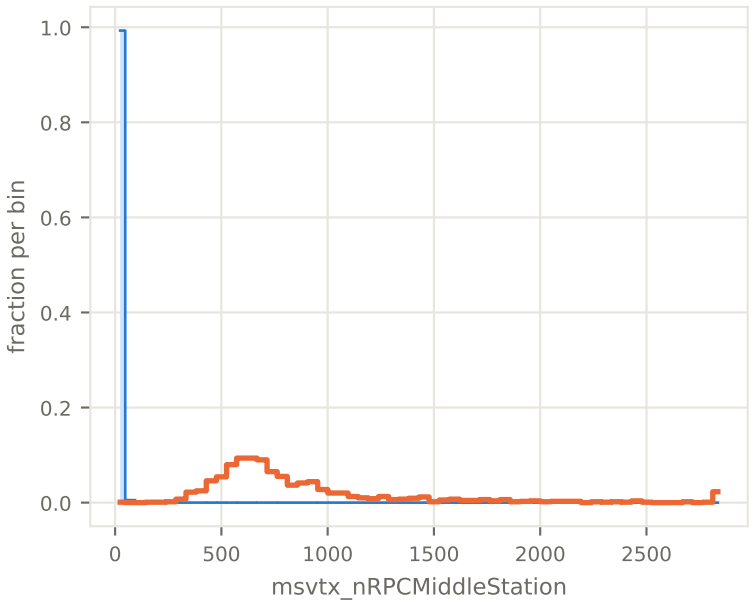
BARREL vs ENDCAP · shape · rank 12 / 98 by |D_barrel - D_endcap| · barrel only

shape D(barrel vs endcap): background 0.998  signal 0.980  separation D: endcap 0.119  barrel 0.520

1a · background (data) · raw

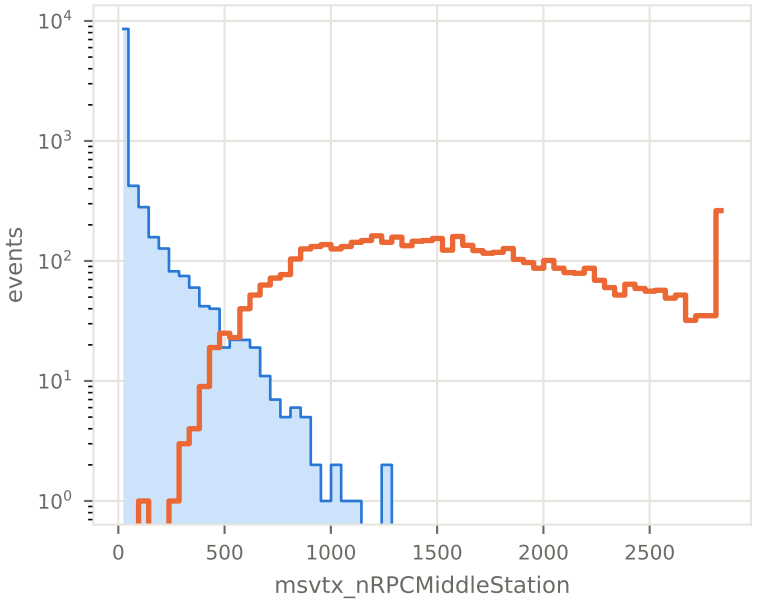



1b · background (data) · normalized

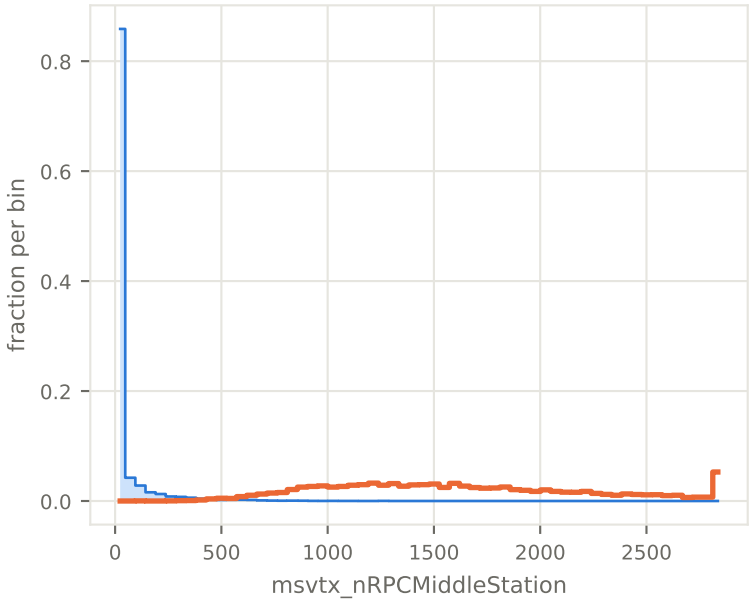



1 · background (data)
shape D = 0.998
endcap n 34,970
barrel n 1,089

2a · combined signal (MC, pooled) · raw




2b · combined signal (MC, pooled) · normalized




2 · combined signal (MC, pooled)
shape D = 0.980
endcap n 9,996
barrel n 4,987

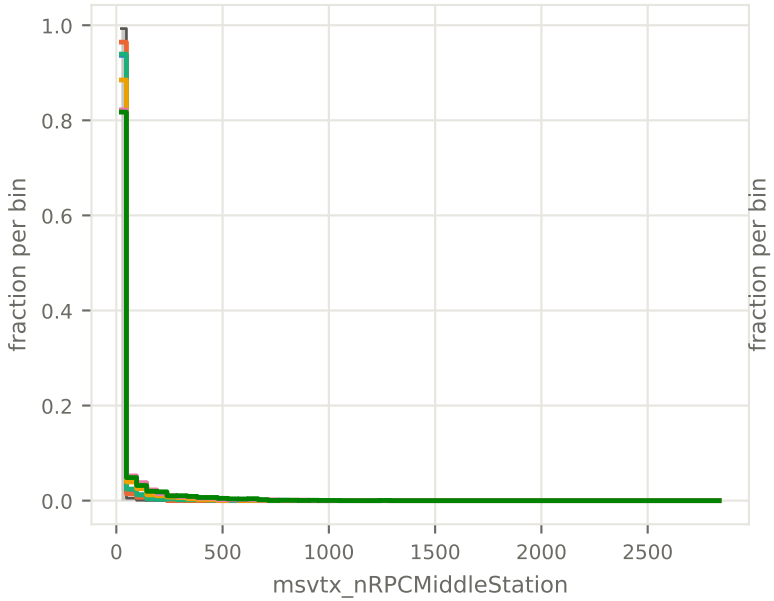
a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_nRPCMiddleStation

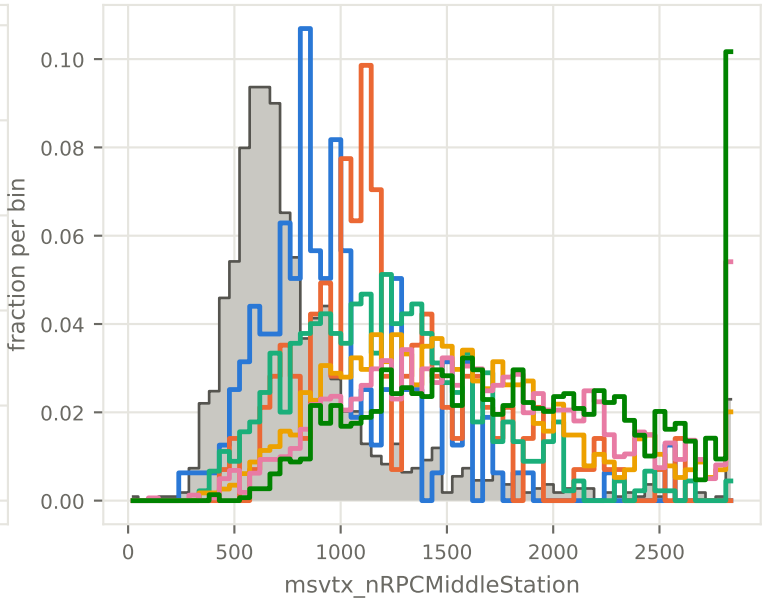
BARREL vs ENDCAP · separation side by side · rank 12 / 98 · barrel only

D_equal  endcap 0.119   barrel 0.520   difference (barrel - endcap) +0.401   (normalized only: raw counts differ ~30x between regions)

1a · ENDCAP · each mass point vs background

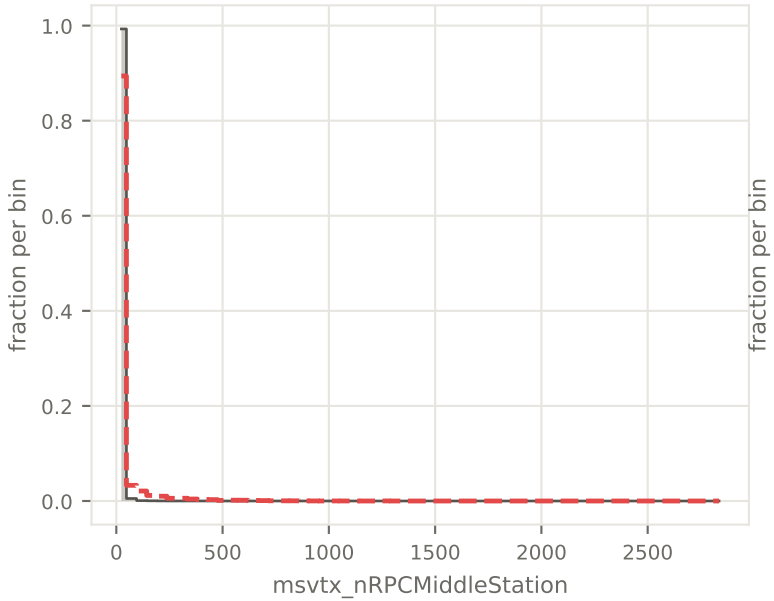



1b · BARREL · each mass point vs background

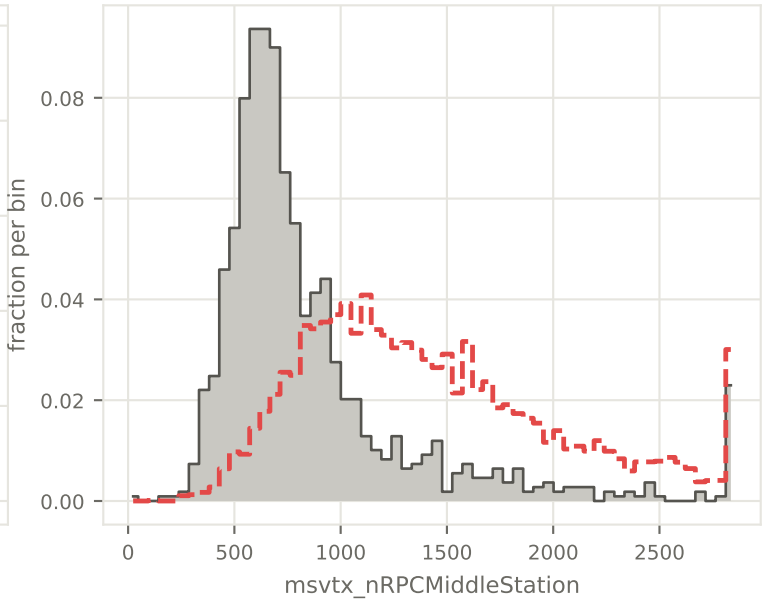



1 · each mass point
D: endcap | barrel
background  n 34,970 | 1,089
mH125_mS15  D 0.067 | 0.314
mH125_mS55  D 0.047 | 0.528
mH200_mS50  D 0.067 | 0.469
mH400_mS100  D 0.129 | 0.605
mH600_mS150  D 0.217 | 0.643
mH1000_mS275  D 0.207 | 0.691

2a · ENDCAP · combined signal vs background




2b · BARREL · combined signal vs background




2 · combined signal
D: endcap | barrel
background  n 34,970 | 1,089
combined, equal weight  D 0.119 | 0.520

a = endcap  b = barrel · normalized · edge bins include under/overflow · NaN and -999 removed